

Modeling with Spotify Data

By: Olivia Pan & Jamie Ngo



01 - INTRODUCTION



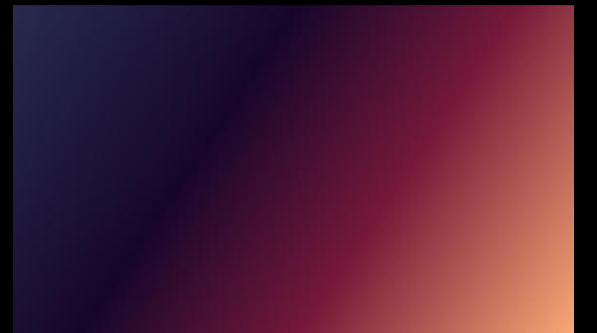
02 - VISUALIZE THE DATA



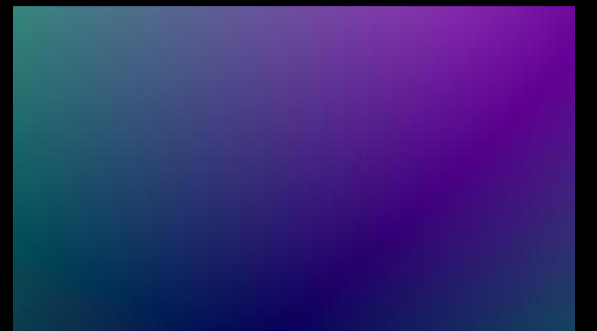
03 - METHOD



04 - RESULTS



05 - CONCLUSION



INTRODUCTION



Dataset Introduction

[Spotify Dataset in Kaggle](#)

Audio Features	
Feature	Description
Energy	A measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy.
Tempo	The speed of a track, measured in beats per minute (BPM).
Danceability	A score describing how suitable a track is for dancing based on tempo, rhythm stability, beat strength and overall regularity.
Loudness	The overall loudness of a track in decibels (dB). Higher values indicate louder tracks overall.
Liveness	The likelihood of a track being performed live. Higher values suggest more audience presence.
Valence	The overall musical positiveness(emotion) of a track. High valence sounds happy; low valence sounds sad or angry.
Speechiness	Measures the presence of spoken words.
Instrumentalness	The likelihood a track contains no vocals. Values closer to 1.0 suggest solely instrumental tracks.
Mode	Indicates the modality of the track.
Key	The musical key, represented as an integer from 0 to 11, mapping to standard Pitch class notation.
Duration_ms	The length of the track in milliseconds.
Acousticness	A confidence measure of whether a track is acoustic(1) or not(0).

The dataset is collated from Spotify's API using two separate python scripts to extract popular and non-popular songs and their associated audio and descriptive features.

Descriptive Features	
Feature	Description
Track Name	The name of the track.
Track Artist	The artist(s) performing the track.
Track Album Name	The album in which the track is featured.
Track Album Release Date	The release date of the album containing the track.
Track ID	A unique identifier assigned to the track by Spotify.
Track Album ID	A unique identifier for the album.
Playlist Name	The name of the playlist where the track is included.
Playlist Genre	The main genre associated with the playlist (e.g., pop, rock, classical).
Playlist Subgenre	A more specific subgenre tied to the playlist (e.g., indie pop, punk rock).
Playlist ID	A unique identifier for the playlist.
Track Popularity	A score (0–100) which is calculated based on total number of streams in relation to other songs.

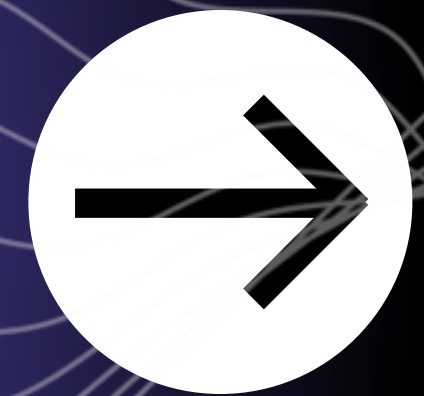
Project Question

What features are determinant for predicting track popularity of a song?

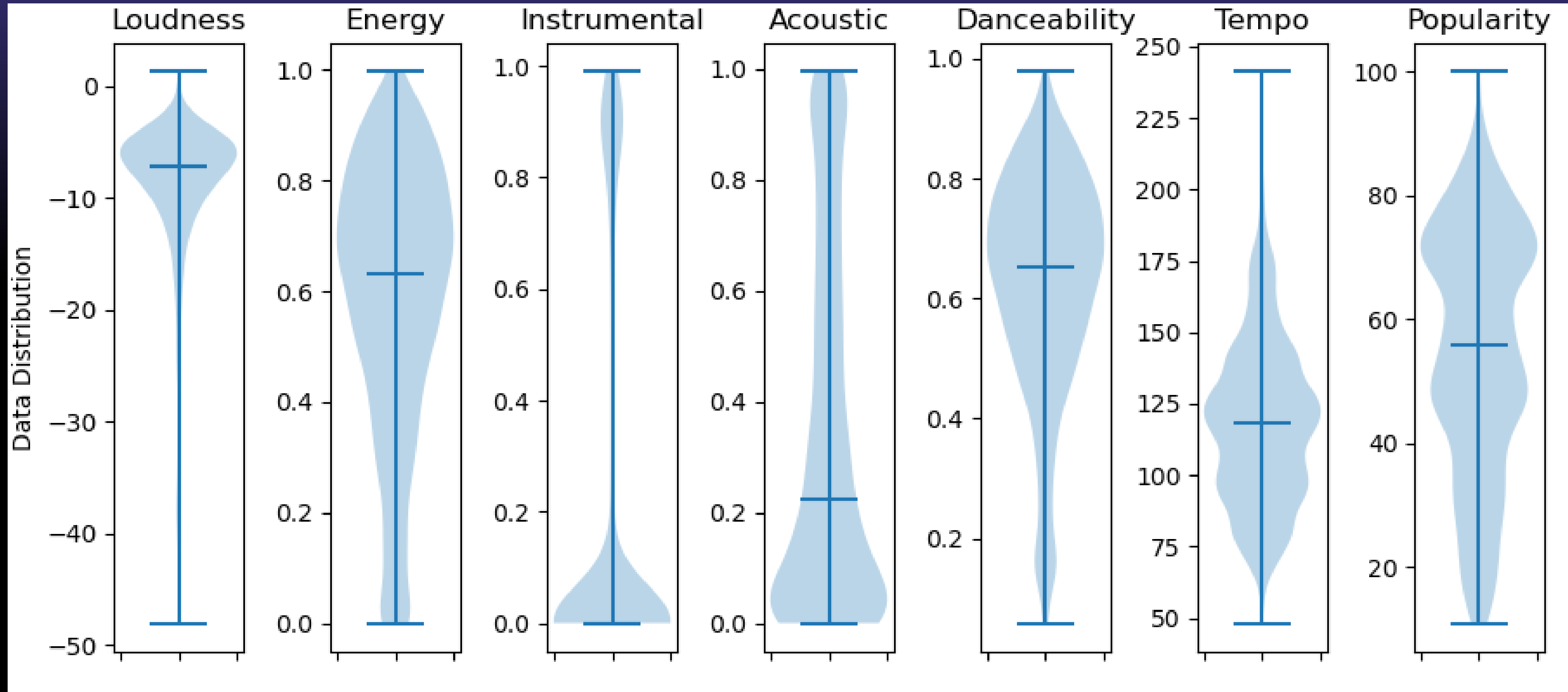
Our expectation

We do expect genre and artist to be influential for music popularity prediction, but we do not expect the regression after feature selection to work well ($R^2 > 0.8$) because music popularity is hard to predict.

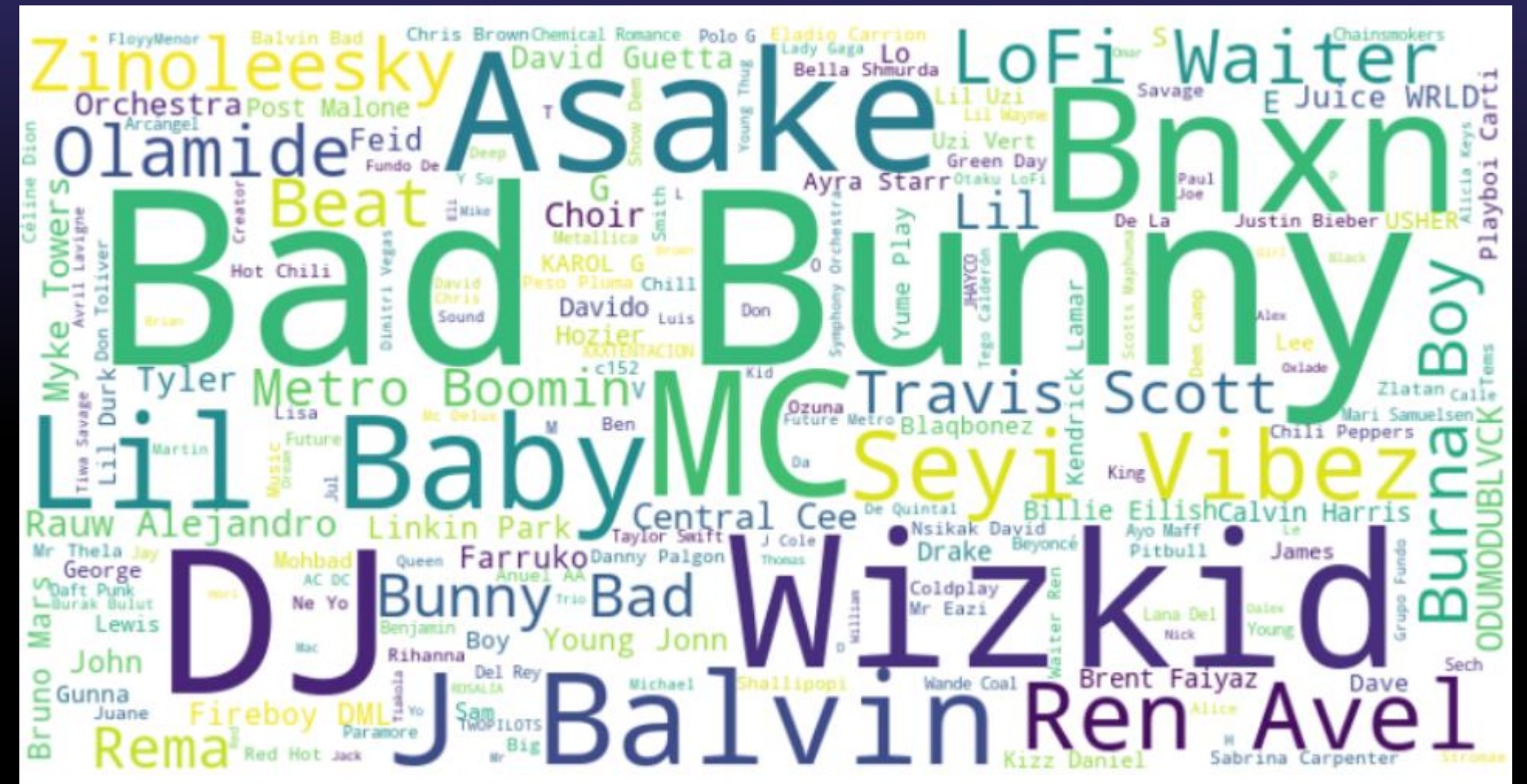
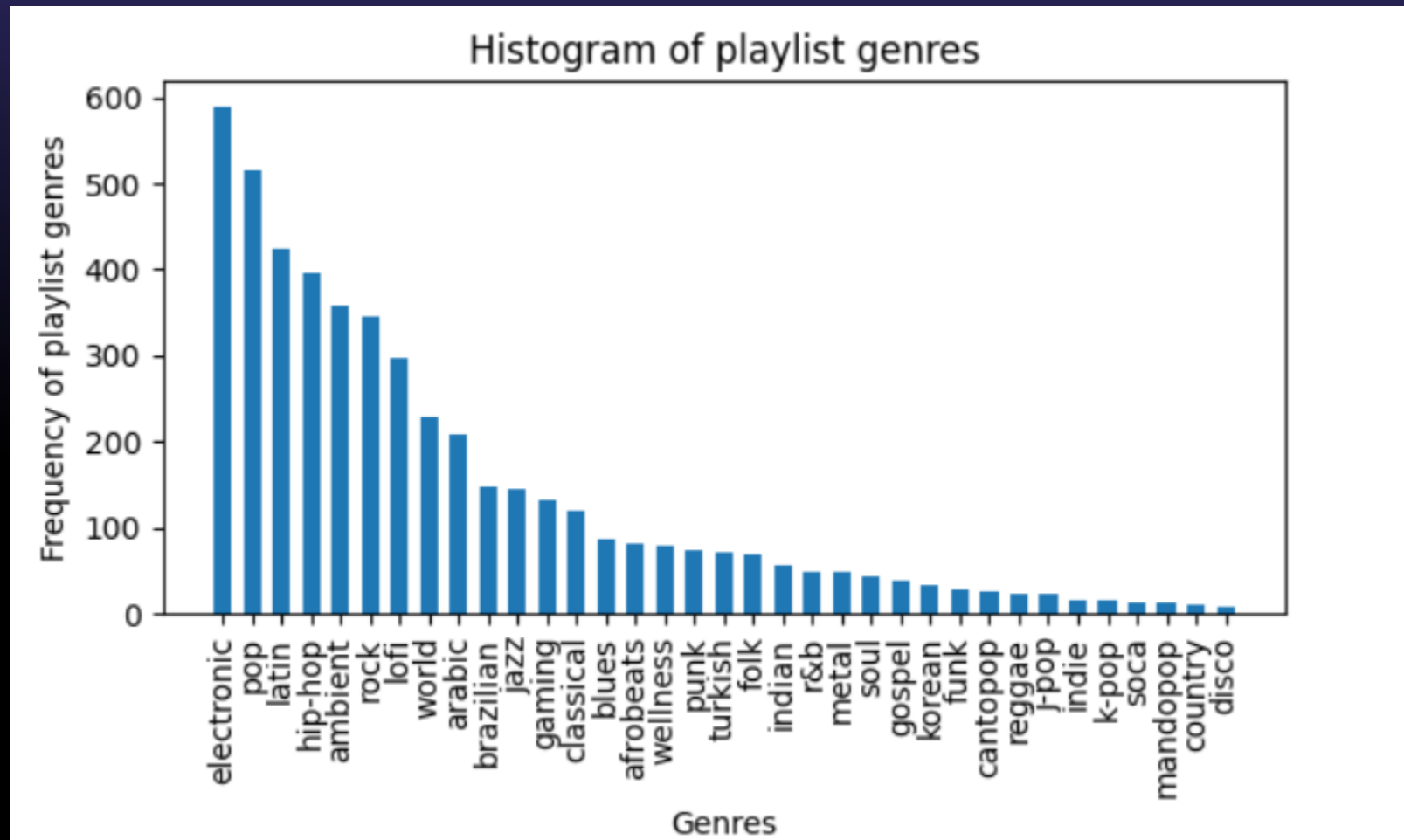
VISUALIZE THE DATA



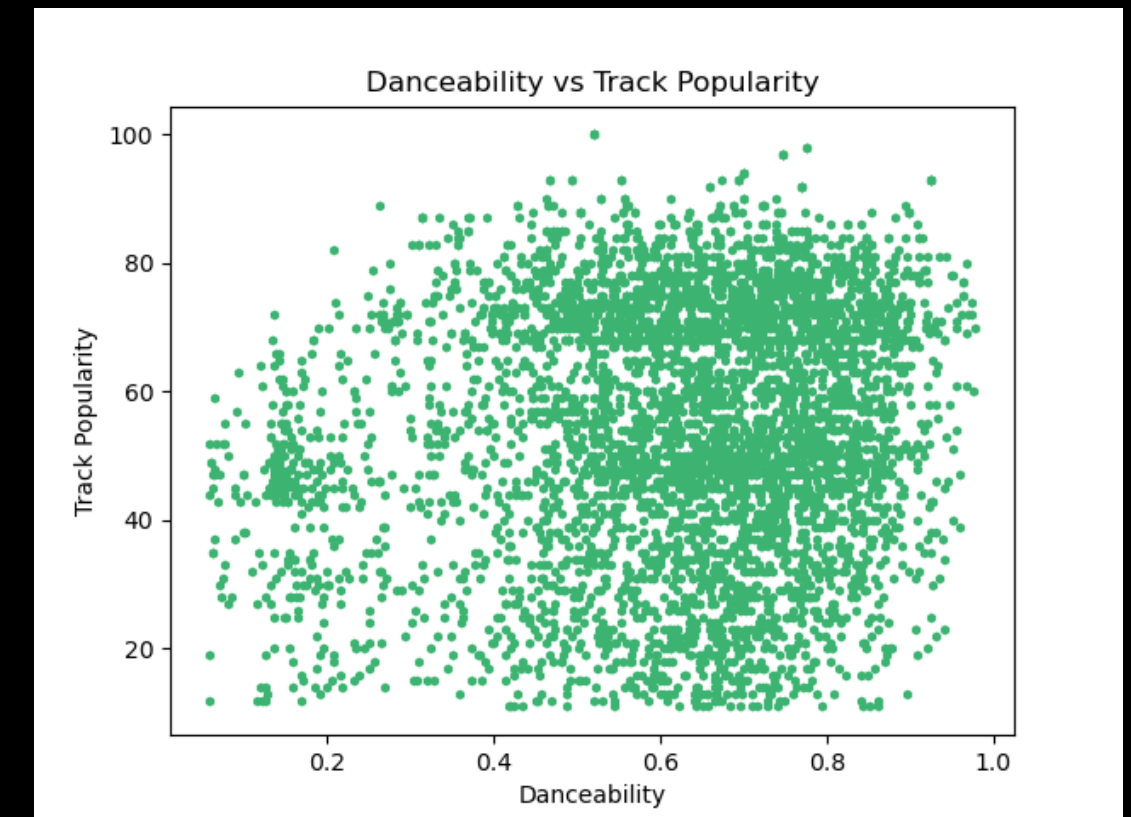
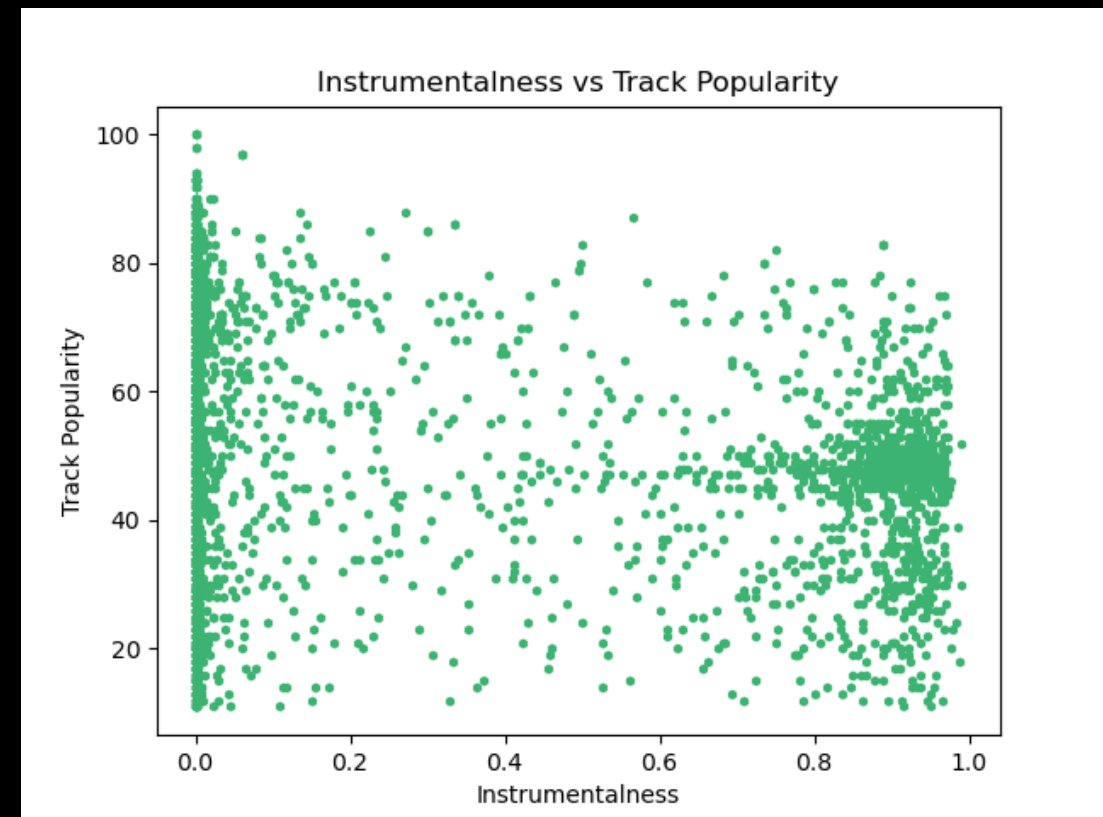
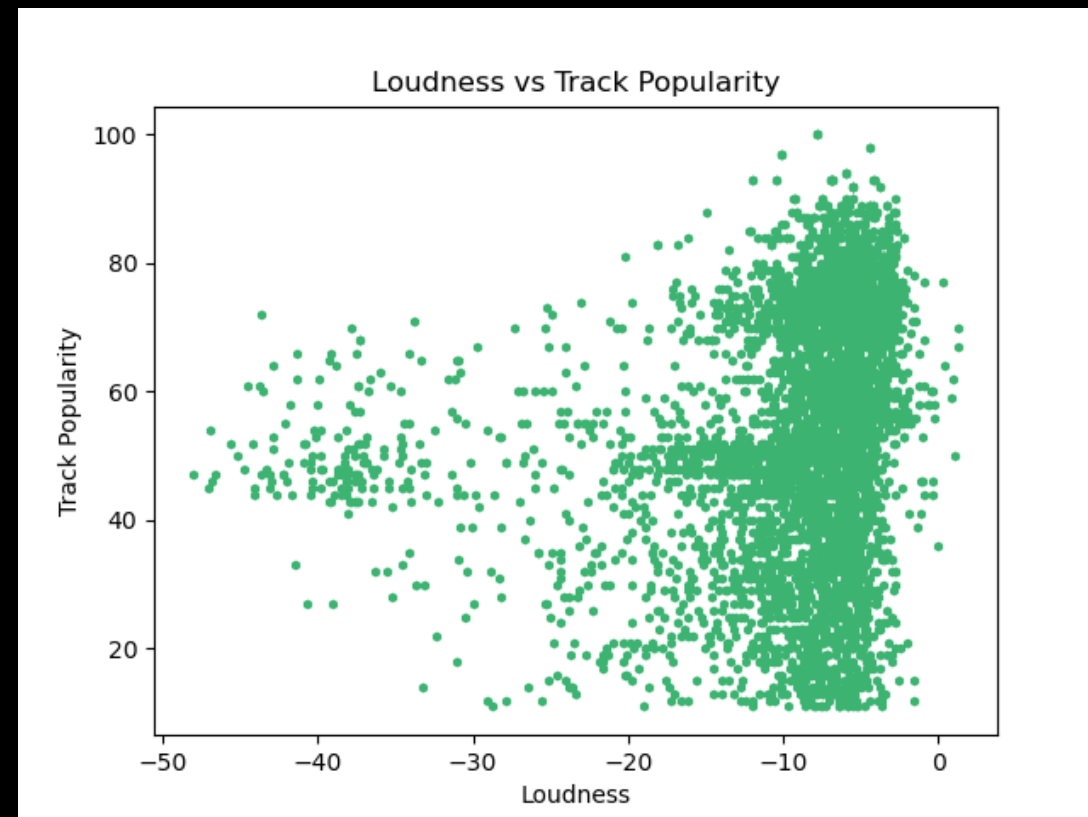
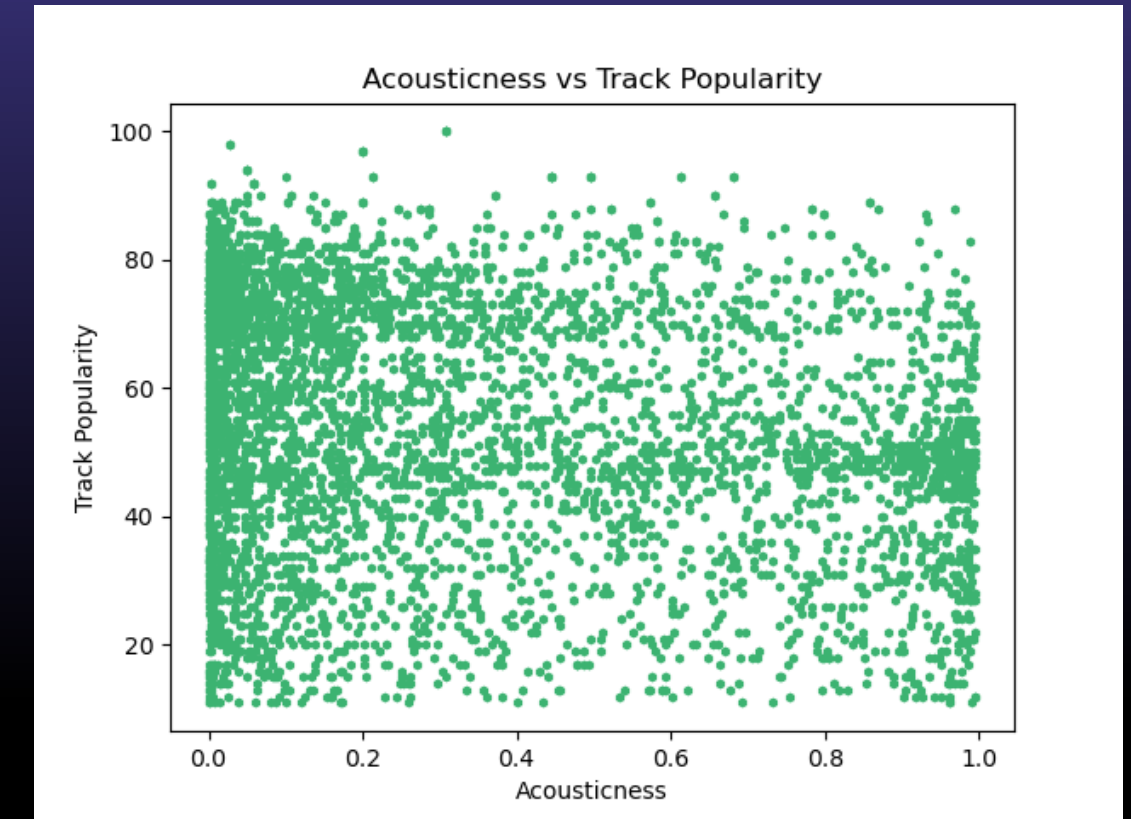
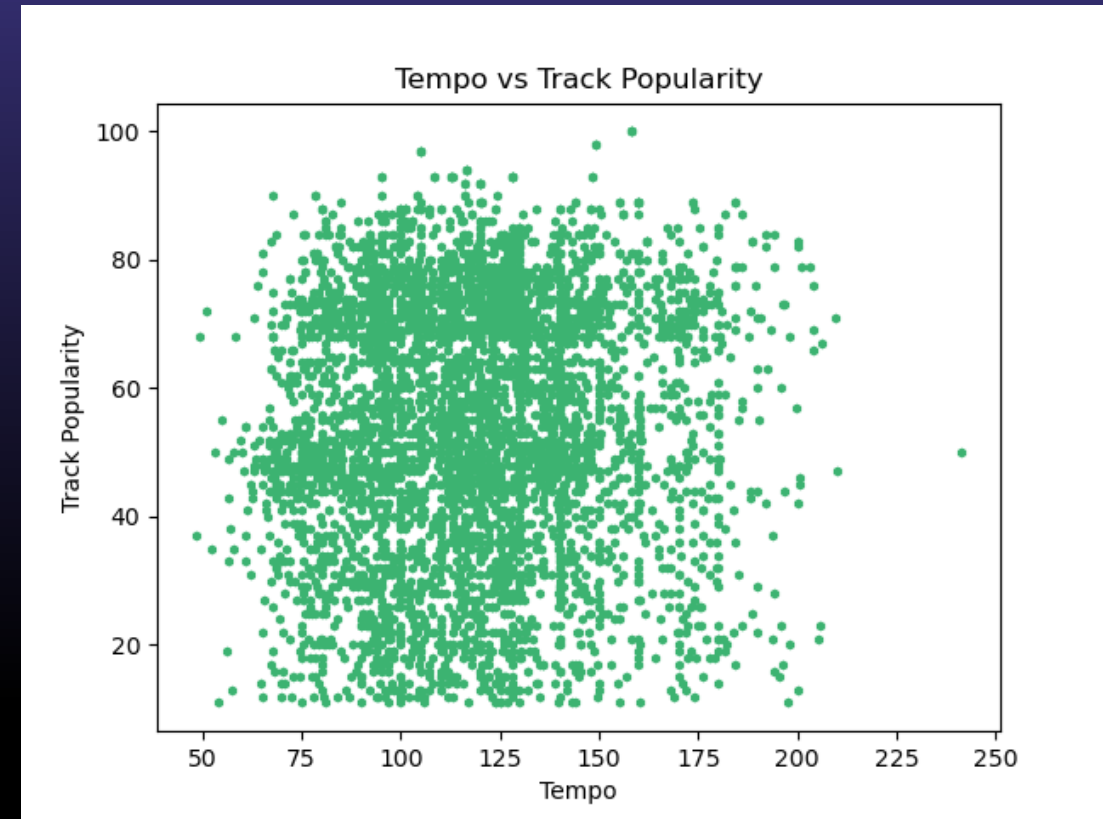
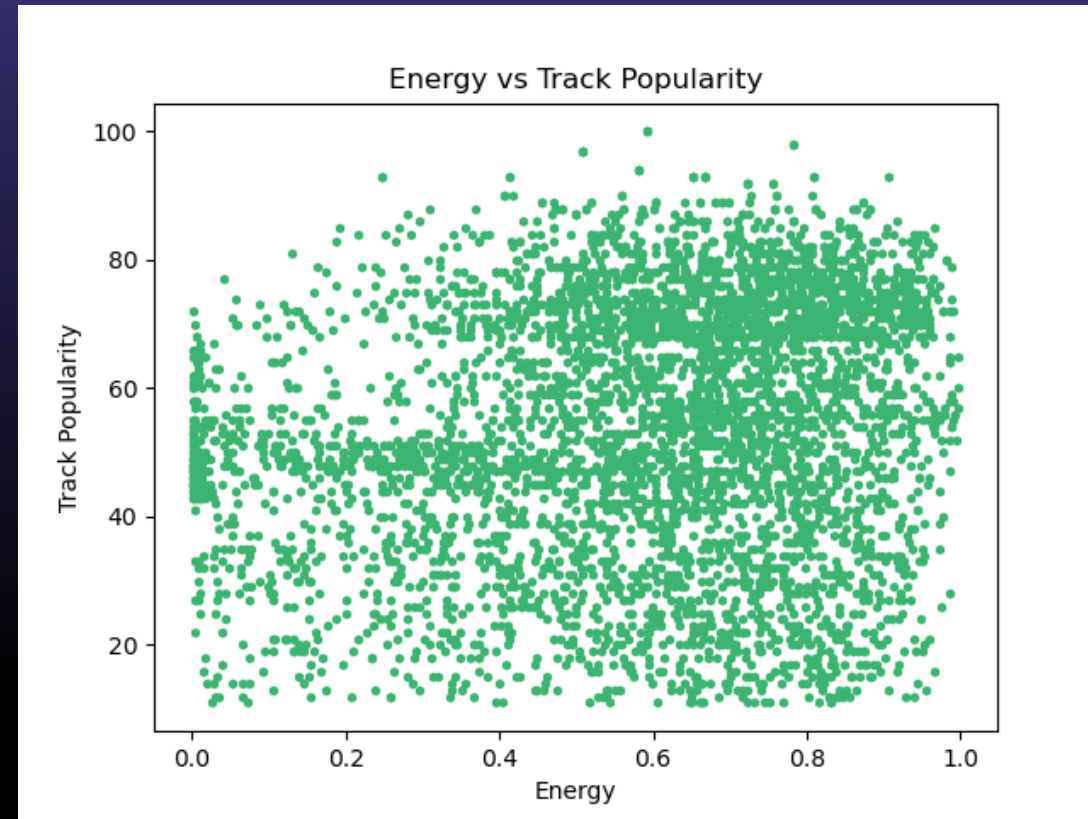
Data Distribution



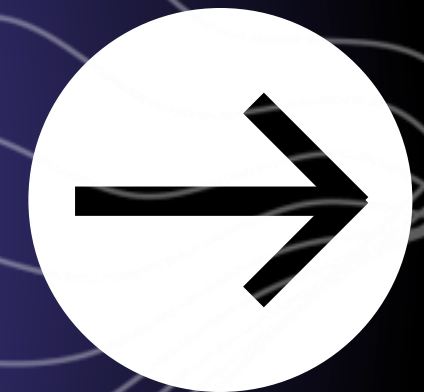
Data Distribution



Feature VS Target



METHOD



Mutual_info_regression

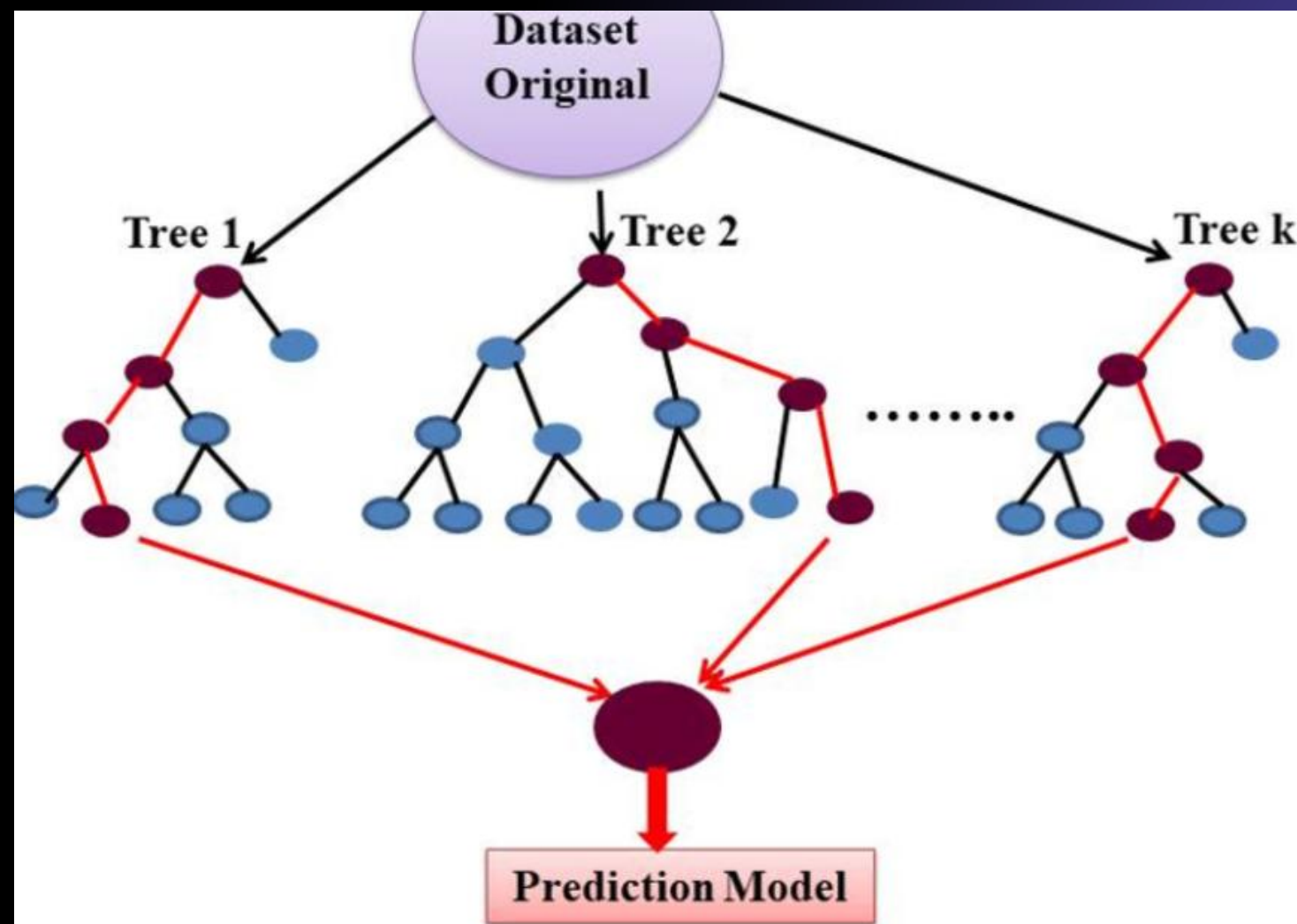
1. Mutual information (MI) measure how much information one variable provides about another: for a feature X and target Y, MI quantifies the reduction in uncertainty of Y given X
2. Model-agnostic: does not assume a specific model, so is useful for preprocessing before modeling
3. Is able to detect non-linear relationships between features and target
4. Robust to monotonic transformations
5. Downside is it looks at each feature independently, ignoring feature interactions



	Feature	Score
7	playlist_subgenre	0.523956
6	playlist_genre	0.413129
11	track_album_release_date	0.254137
3	track_artist	0.214635
18	loudness	0.176177
13	instrumentalness	0.173398
5	energy	0.158753
19	acousticness	0.152599
2	danceability	0.141180
4	duration_ms	0.140833
17	tempo	0.115972
20	liveness	0.115371
15	valence	0.107534
1	speechiness	0.103360
12	id	0.096127
8	track_href	0.096102
14	track_id	0.095846
16	key	0.031110
0	time_signature	0.026771
9	mode	0.009057
10	type	0.002129

1. Extra Trees is an ensemble method similar to Random Forest, but with more randomness
2. Chooses random split among a random subset of features, resulting in lower variance
3. More suitable for noisy dataset and high-dimensional data

Extra Trees Regressor



PROCESS

1. Download dataset

Obtained the spotify dataset from Kaggle

- high popularity
- low popularity

2. Data wrangling

Combined the datasets and cleaned it up

- drop 'na' rows
- drop unwanted columns like playlist name and album name
- encode categorical columns

3. Feature selection

Ran mutual_info_regression

- playlist_genre, artist, loudness, energy, instrumentalness, acousticness, danceability, tempo

4. Graphing

Visualize the dataset features

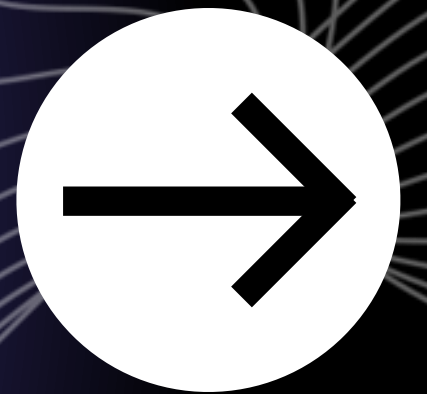
- Violin plots
- Scatter plots
- Bar charts
- Word cloud

5. Regression

Ran ExtraTreesRegressor

- plot the predictions and results

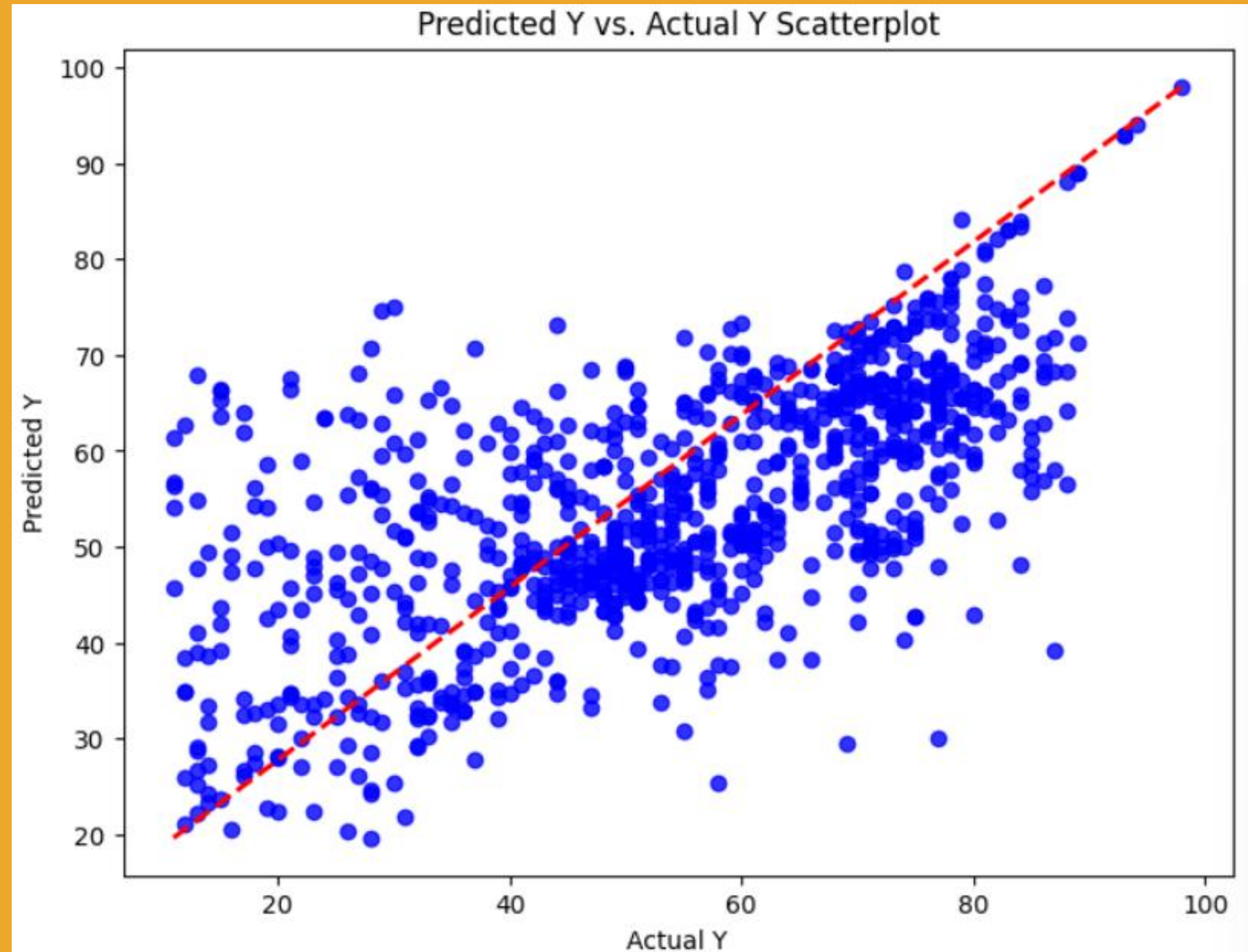
RESULTS

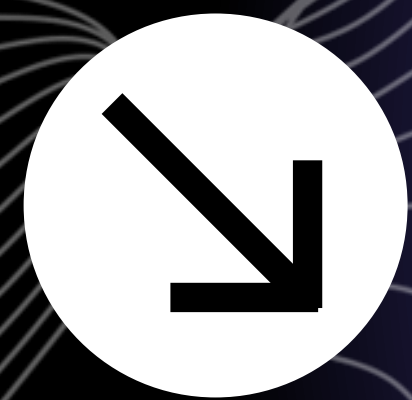


ExtraTreesRegressor results

RMSE:
222.78

R^2 Score:
0.427





CONCLUSION



- Strongest factors were genre and artist
- Music is diverse and difficult to predict popularity
- Many factors contribute to popularity
- Need to think about current trends
- playlist_genre, artist, loudness, energy, instrumentality, acousticness, danceability, tempo are able to capture some trend in track_popularity with some limitations

Potential improvements:

1. group songs by genres (controlled for genre variable) or artists
2. Test out other datasets with different variables

References

Ameh, S. (2025, January 7). *Spotify Music dataset*. Kaggle.
<https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset?resource=download>

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.ExtraTreesRegressor.html>

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.mutual_info_regression.html